Sazid Rahman Simanto

Robotics Software Engineer — ROS2, SLAM, Sensor Fusion, LiDAR/Radar/Camera Perception

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Professional Summary

Robotics Software Engineer with experience in mobile robotics perception, SLAM/odometry, navigation, simulation, and multi-sensor fusion. Hands-on with ROS2, C++, Python, Linux, Jetson Orin NX, 3D LiDAR, FMCW radar, stereo depth cameras, and ToF cameras. Experienced in developing robotics and perception pipelines in simulation and on real hardware, with prior production software experience at Samsung and automotive LiDAR software experience at Ibeo.

Core Skills

- **Robotics:** ROS2, SLAM, Odometry, Navigation, Sensor Fusion, Camera Calibration
- **Perception & AI:** Object Detection, Segmentation, Radar Signal Processing, Point Cloud Processing, PyTorch, OpenCV, Open3D, CUDA
- **Programming:** C++, C, Python, MATLAB, Swift
- **Hardware:** Jetson Orin NX, Livox LiDAR, RealSense Stereo Camera, TI AWR1843 FMCW Radar, ToF Camera, RS485 Motors
- **Tools:** Linux, Git, Docker, CMake, Jenkins, Qt, Confluence
- **Simulation:** NVIDIA Isaac Sim, Unity
- **Languages:** English C1, German A1, Bengali Native

Professional Experience

Research Associate – Robotics Perception May 2025 – Present
Institute of Computer Engineering, University of Lübeck, Germany

- Developing perception and localization modules for an autonomous mobile robot platform in cleanroom environments as part of the EU-funded AIMS5.0 Industry 5.0 initiative.
- Developed SLAM and odometry pipelines for map generation and robot state estimation.
- Built sensor-fusion modules combining radar and ToF depth-camera data for perception and state estimation.
- Integrated multi-sensor robotics hardware including 3D LiDAR, FMCW radar, ToF camera, and depth camera with embedded compute.
- Worked with ROS2, C++, Python, OpenCV, Docker, and Jetson-based deployment workflows.

Student Research Assistant – Autonomous Systems Nov 2022 – Feb 2025
Institute of Computer Engineering, University of Lübeck, Germany

- Contributed to the BMBF-funded MANNHEIM-EMDRIVE project on energy-efficient computing architectures for autonomous and connected vehicles.
- Designed and optimized simulation data-flow components for AI accelerator architectures.
- Developed near-sensor radar data processing pipelines for perception and object tracking.

C++ Software Engineer Aug 2022 – Oct 2022
Ibeo Automotive Systems GmbH, Hamburg, Germany

- Contributed to C++ software development in an automotive LiDAR perception environment.
- Worked with production-style C++ code, debugging, integration, and version-control workflows.
- Gained practical experience in automotive perception software and LiDAR-based autonomous driving systems.

Software Engineer Oct 2019 – Mar 2021
Samsung R&D Institute, Dhaka, Bangladesh

- Developed and refactored features for the Samsung SmartThings iOS application.
- Worked in a production software team using Git, Jenkins, code review, and release workflows.
- Contributed to an IoT platform application with a 4.6★ App Store rating and top-15 Lifestyle category ranking.

Selected Projects

SLIM LIO – Lightweight LiDAR-Inertial Odometry GitHub

- Built a lightweight LiDAR-inertial odometry pipeline in C++/ROS2.
- Implemented tightly coupled IEKF-based state estimation and point-to-plane scan matching.

Autonomous Mobile Robot Platform GitHub

- Developing a Jetson Orin NX-based mobile robot platform with Livox LiDAR, RealSense stereo camera, TI AWR1843 radar, and RS485 smart motors.
- Designing fused power distribution, remote operation, motor control, and live sensor visualization for robotics experiments.

Early-Level Sensor Fusion on Jetson Orin NX Demo Video

- Combined RGB, ToF point-cloud, and radar data for real-time embedded perception.
- Achieved 11 FPS real-time fusion on Jetson Orin NX.

AIMS5.0 – Autonomous Mobile Robot for Cleanroom Environments Project Website

- Developed SLAM, odometry, and sensor-fusion modules for an EU-funded Industry 5.0 robotics project.

MANNHEIM-EMDRIVE – Radar Data Processing and AI Accelerator Simulation Project Website

- Developed radar data processing pipelines and simulation data-flow components for autonomous vehicle computing research.

Education

M.Sc. Robotics and Autonomous Systems 2021 – 2025
University of Lübeck, Germany

B.Sc. Computer Science & Engineering 2015 – 2019
North South University, Bangladesh

Certifications & Activities

- Coursera Deep Learning Specialization
- ACM ICPC Dhaka Regional Participant: 2015, 2018, 2019
- Competitive Programming: Codeforces Specialist, CodeChef 4★, HackerRank Silver